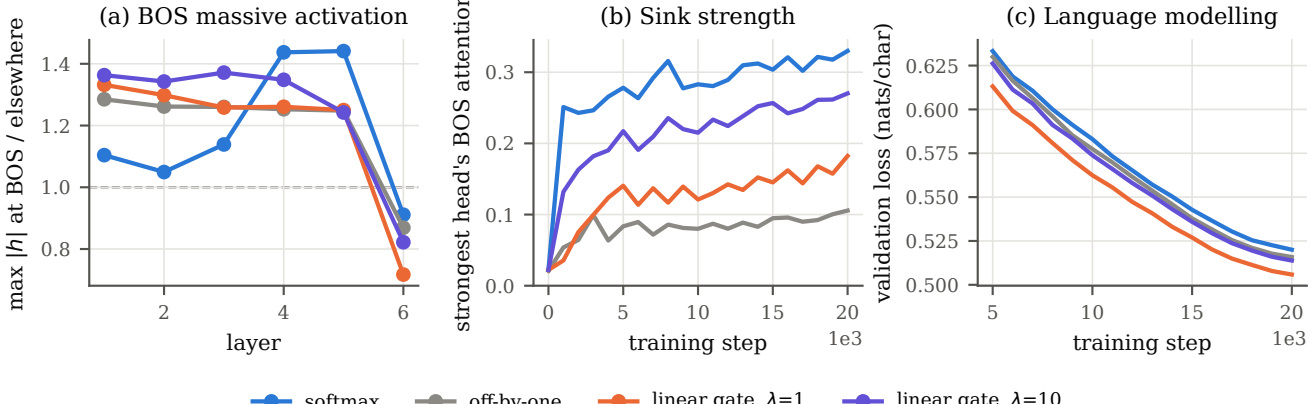
(a) BOS massive activation
max |h| at BOS / elsewhere
1.4
1.2
1.0
0.8
2
4
6
layer
(b) Sink strength
strongest head's BOS attention
0.3
0.2
0.1
0.0
0
5
10
15
20
1e3
training step
(c) Language modelling
validation loss (nats/char)
0.625
0.600
0.575
0.550
0.525
0.500
5
10
15
20
1e3
training step
softmax
off-by-one
linear gate, λ=1
linear gate, λ=10